

11th grade

Term 3 — Lesson 1 Practice Activity (C1–C3) with Solutions

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Evaluated criteria

- C1: Describe the concept of a statistical hypothesis in context.
- C2: Identify the null and alternative hypotheses using correct parameter notation.
- C3: Determine whether a test is one tailed or two tailed based on the claim.

Introduction

General problem

A benchmark claim may refer either to a population proportion or to a population mean.

For a proportion setting, the claim states that specific percentage (for example, 62%) of the population has the characteristic of interest. A sample of number of samples (for example, 200) observations gives number of success observed successes (for example, 112). Use a significance level of significance percentage (for example, 5%).

For a mean setting, the claim states that the population mean is a benchmark mean value (for example, 75). A sample of number of samples (for example, 36) observations gives a sample mean value (for

example, 79). If needed, the population standard deviation is given as a known value (for example, 12). Use a significance level of significance percentage (for example, 1%).

Questions/Tasks.

- (C1) Identify whether the setting is a proportion claim or a mean claim, and map the benchmark value, sample information, and significance level to the correct symbols.
- (C2) Write the null and alternative hypotheses for the three directions of the claim.
- (a) Write the null hypothesis as “parameter equals the benchmark” and the alternative as “parameter is lower than the benchmark”.
 - (b) Write the null hypothesis as “parameter equals the benchmark” and the alternative as “parameter is higher than the benchmark”.
 - (c) Write the null hypothesis as “parameter equals the benchmark” and the alternative as “parameter is different from the benchmark”.
- (C3) Classify the tail type for each direction of the claim and justify it in words.

C1

- Identify whether the problem is about a population proportion or a population mean.
- For a proportion problem:
 - state that p is the true population proportion,
 - state that p_0 is the benchmark value (for example, 62%),
 - define n as the sample size (for example, 200),
 - define x as the observed count (for example, 112),
 - define α as the significance level (for example, 5%).
- For a mean problem:
 - state that μ is the true population mean,
 - state that μ_0 is the benchmark value (for example, 75),
 - define n as the sample size (for example, 36),
 - define $\bar{x}$ as the sample mean (for example, 79),
 - define σ as the known population standard deviation when it is provided (for example, 12),
 - define α as the significance level (for example, 1%).
- Match each symbol correctly to the numerical values given in the statement.
- Parts “lower”, “higher”, and “different” use the same parameter and benchmark, but different alternative claims.

C2

- Write the hypotheses correctly for each part (a), (b), and (c), and include α in each case.

- For a proportion problem, use p and p_0 :
 - (a) $H_0: p = p_0$, $H_a: p < p_0$, with the stated significance level. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is lower than the benchmark.
 - (b) $H_0: p = p_0$, $H_a: p > p_0$, with the stated significance level. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is higher than the benchmark.
 - (c) $H_0: p = p_0$, $H_a: p \neq p_0$, with the stated significance level. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is different from the benchmark.
- For a mean problem, use μ and μ_0 :
 - (a) $H_0: \mu = \mu_0$, $H_a: \mu < \mu_0$, with the stated significance level. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is lower than the benchmark.
 - (b) $H_0: \mu = \mu_0$, $H_a: \mu > \mu_0$, with the stated significance level. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is higher than the benchmark.
 - (c) $H_0: \mu = \mu_0$, $H_a: \mu \neq \mu_0$, with the stated significance level. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is different from the benchmark.
- In every case, keep the null hypothesis as “parameter equals the benchmark” and match the inequality direction to lower, higher, or different.

C3

- Classify the tail type for each direction of the claim and justify it in words.
 - (a) left-tailed test, because the claim says “lower”.



- (b) right-tailed test, because the claim says “higher”.



- (c) two-tailed test, because the claim says “different”.



- This mapping is the same for both proportion and mean tests.

P1 1 Daily app check-ins for students and recent graduates

A university career center uses a mobile app to support students and recent graduates with internship alerts, interview reminders, and short productivity check-ins. The center is reviewing whether users are staying engaged enough for the app to justify next semester's funding, especially because many users are balancing classes, part-time work, and job applications. The current benchmark says that 62% of active users complete the daily check-in. A class project reviews 200 active users and finds that 112 completed the daily check-in. They will use a 5% significance level for this check.

Questions/Tasks.

- (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value, the sample size, and the observed sample result for the daily check-in proportion to the symbols used in the solution.
- (C2) Write hypotheses for each case using the stated significance level.
- (a) Using a significance level of 5%, write the null and alternative hypotheses for the case where the true daily check-in proportion is lower than 62%.
 - (b) Using a significance level of 5%, write the null and alternative hypotheses for the case where the true daily check-in proportion is higher than 62%.
 - (c) Using a significance level of 5%, write the null and alternative hypotheses for the case where the true daily check-in proportion is different from 62%.
- (C3) For each case, classify the test as left-tailed, right-tailed, or two-tailed and justify using the wording of the alternative claim.

C1

- This is a population proportion setting.
- Let p be the true population proportion of active users who check in daily.
- Let p_0 be the benchmark proportion, with $p_0 = 62\%$.
- Let n be the sample size, with $n = 200$.
- Let x be the observed number of daily check-ins, with $x = 112$.
- Let α be the significance level, with $\alpha = 5\%$.
- Parts “lower”, “higher”, and “different” use the same parameter p and the same benchmark p_0 , but different alternative claims.

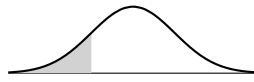
C2

- (a) $H_0: p = 0.62$, $H_a: p < 0.62$, $\alpha = 0.05$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is lower than the benchmark.
- (b) $H_0: p = 0.62$, $H_a: p > 0.62$, $\alpha = 0.05$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is higher than the benchmark.

(c) $H_0: p = 0.62$, $H_a: p \neq 0.62$, $\alpha = 0.05$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is different from the benchmark.

C3

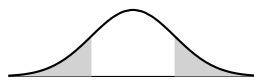
(a) Left-tailed test because the alternative claims the parameter is lower than the benchmark.



(b) Right-tailed test because the alternative claims the parameter is higher than the benchmark.



(c) Two-tailed test because the alternative claims the parameter is different from the benchmark.



P2 2 Evening study time during a demanding term

A university academic support office is evaluating whether students are spending enough time studying each evening during a demanding midterm period. Many of them are combining coursework with internships, long commutes, or part-time jobs, so the office wants to know whether the reported average still reflects current study habits. The report claims that students study an average of 75 minutes per evening. A sample of 36 students has an average of 79 minutes per evening. Assume the population standard deviation is known to be 12 minutes. They will use a 10% significance level for this check.

Questions/Tasks.

- (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value, the sample size, and the observed sample result for the mean study time to the symbols used in the solution.
- (C2) Write hypotheses for each case using the stated significance level.
- (a) Using a significance level of 10%, write the null and alternative hypotheses for the case where the true mean study time is lower than 75 minutes.
 - (b) Using a significance level of 10%, write the null and alternative hypotheses for the case where the true mean study time is higher than 75 minutes.
 - (c) Using a significance level of 10%, write the null and alternative hypotheses for the case where the true mean study time is different from 75 minutes.
- (C3) For each case, classify the test as left-tailed, right-tailed, or two-tailed and justify using the wording of the alternative claim.

C1

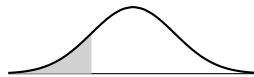
- This is a population mean setting.
- Let μ be the true population mean study time per evening for students.
- Let μ_0 be the benchmark mean, with $\mu_0 = 75$.
- Let n be the sample size, with $n = 36$.
- Let $\bar{x}$ be the sample mean study time, with $\bar{x} = 79$.
- Let σ be the known population standard deviation, with $\sigma = 12$.
- Let α be the significance level, with $\alpha = 10\%$.
- Parts “lower”, “higher”, and “different” use the same parameter μ and the same benchmark μ_0 , but different alternative claims.

C2

- (a) $H_0: \mu = 75$, $H_a: \mu < 75$, $\alpha = 0.10$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is lower than the benchmark.
- (b) $H_0: \mu = 75$, $H_a: \mu > 75$, $\alpha = 0.10$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is higher than the benchmark.
- (c) $H_0: \mu = 75$, $H_a: \mu \neq 75$, $\alpha = 0.10$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is different from the benchmark.

C3

- (a) Left-tailed test because the alternative claims the parameter is lower than the benchmark.



- (b) Right-tailed test because the alternative claims the parameter is higher than the benchmark.



- (c) Two-tailed test because the alternative claims the parameter is different from the benchmark.



P3 3 Defect rate in portable chargers

A consumer electronics brand markets portable chargers to people who depend on their phones for maps, payment apps, job communication, and ride-hailing. After several customer complaints from users who said the chargers failed during commutes or long campus and work days, the quality team reviews whether the production line is still meeting its usual standard. The factory process has a

benchmark defect proportion of 6%. In a quality sample of 400 chargers, 34 are defective. Because defect monitoring requires a stricter standard, they will use a 2% significance level for this check.

Questions/Tasks.

- (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value, the sample size, and the observed sample result for the defect proportion to the symbols used in the solution.
- (C2) Write hypotheses for each case using the stated significance level.
- (a) Using a significance level of 2%, write the null and alternative hypotheses for the case where the true defect proportion is lower than 6%.
 - (b) Using a significance level of 2%, write the null and alternative hypotheses for the case where the true defect proportion is higher than 6%.
 - (c) Using a significance level of 2%, write the null and alternative hypotheses for the case where the true defect proportion is different from 6%.
- (C3) For each case, classify the test as left-tailed, right-tailed, or two-tailed and justify using the wording of the alternative claim.

C1

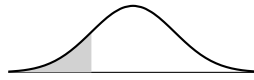
- This is a population proportion setting.
- Let p be the true population proportion of defective chargers from the process.
- Let p_0 be the benchmark defect proportion, with $p_0 = 6\%$.
- Let n be the sample size, with $n = 400$.
- Let x be the observed number of defective chargers, with $x = 34$.
- Let α be the significance level, with $\alpha = 2\%$.
- Parts “lower”, “higher”, and “different” use the same parameter p and the same benchmark p_0 , but different alternative claims.

C2

- (a)** $H_0: p = 0.06$, $H_a: p < 0.06$, $\alpha = 0.02$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is lower than the benchmark.
- (b)** $H_0: p = 0.06$, $H_a: p > 0.06$, $\alpha = 0.02$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is higher than the benchmark.
- (c)** $H_0: p = 0.06$, $H_a: p \neq 0.06$, $\alpha = 0.02$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is different from the benchmark.

C3

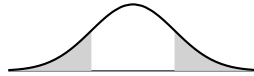
(a) Left-tailed test because the alternative claims the parameter is lower than the benchmark.



(b) Right-tailed test because the alternative claims the parameter is higher than the benchmark.



(c) Two-tailed test because the alternative claims the parameter is different from the benchmark.



P4 4 Homework completion time in an evening certification program

A digital learning platform offers evening certification courses for people trying to strengthen their resumes while working or looking for their first full-time job. Because many users study late at night and have limited free time, the platform team is checking whether homework tasks are taking longer or shorter than expected and whether assignment design should be changed. The platform states that the average homework completion time is 48 minutes. A sample of 49 assignments gives an average of 45 minutes. Assume the population standard deviation is known to be 10 minutes. To be stricter because assignment design changes affect many users, they will use a 1% significance level for this check.

Questions/Tasks.

- (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value, the sample size, and the observed sample result for the mean completion time to the symbols used in the solution.
- (C2) Write hypotheses for each case using the stated significance level.
- (a) Using a significance level of 1%, write the null and alternative hypotheses for the case where the true mean completion time is lower than 48 minutes.
 - (b) Using a significance level of 1%, write the null and alternative hypotheses for the case where the true mean completion time is higher than 48 minutes.
 - (c) Using a significance level of 1%, write the null and alternative hypotheses for the case where the true mean completion time is different from 48 minutes.
- (C3) For each case, classify the test as left-tailed, right-tailed, or two-tailed and justify using the wording of the alternative claim.

C1

- This is a population mean setting.
- Let μ be the true population mean homework completion time for all assignments on the platform.
- Let μ_0 be the benchmark mean, with $\mu_0 = 48$.

- Let n be the sample size, with $n = 49$.
- Let $\bar{x}$ be the sample mean completion time, with $\bar{x} = 45$.
- Let σ be the known population standard deviation, with $\sigma = 10$.
- Let α be the significance level, with $\alpha = 1\%$.
- Parts “lower”, “higher”, and “different” use the same parameter μ and the same benchmark μ_0 , but different alternative claims.

C2

(a) $H_0: \mu = 48, H_a: \mu < 48, \alpha = 0.01$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is lower than the benchmark.

(b) $H_0: \mu = 48, H_a: \mu > 48, \alpha = 0.01$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is higher than the benchmark.

(c) $H_0: \mu = 48, H_a: \mu \neq 48, \alpha = 0.01$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is different from the benchmark.

C3

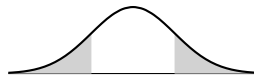
(a) Left-tailed test because the alternative claims the parameter is lower than the benchmark.



(b) Right-tailed test because the alternative claims the parameter is higher than the benchmark.



(c) Two-tailed test because the alternative claims the parameter is different from the benchmark.



P5 5 Secure login success in a budgeting app

A personal budgeting app recently replaced its login flow with a new security verification step. After the update, some users reported both failed attempts and, in other cases, smoother access on newer devices. The product team now wants to check whether the first-attempt login success rate has changed from the historical benchmark, since any shift could affect support costs and user retention. The platform benchmark for first-attempt login success is 91%. In a sample of 320 attempts, 301 are successful. They will use a 5% significance level for this check. (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value of 91%, the sample size, and the observed sample result for the first-attempt login success proportion to the symbols used in the solution. (C2) Using the 5% significance level, write the null and alternative hypotheses for the case where the true first-attempt login success proportion is different from 91%. (C3) Classify the test and justify it in words for the case where the true first-attempt login success proportion is different from 91%.

C1

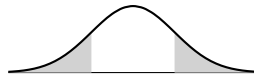
- This is a population proportion setting.
- Let p be the true population proportion of successful first-attempt logins across all attempts.
- Let p_0 be the benchmark success proportion, with $p_0 = 91\%$.
- Let n be the sample size, with $n = 320$.
- Let x be the observed number of successful first-attempt logins, with $x = 301$.
- Let α be the significance level, with $\alpha = 5\%$.
- The alternative claim in this problem is that the proportion is different from the benchmark.

C2

$H_0: p = 0.91$, $H_a: p \neq 0.91$, $\alpha = 0.05$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is different from the benchmark.

C3

Two-tailed test because the alternative claims the parameter is different from the benchmark.



P6 6 Commute time to campus

A district transportation office is monitoring whether a new express-bus plan is meeting its promise to reduce student travel time, especially for students who attend classes while working part time. The plan was designed to keep the average commute at or below the district benchmark of 32 minutes; if the average is not lower, the office may need to redesign routes before the next term. A sample of 40 students gives an average of 34 minutes. Assume the population standard deviation is known to be 8 minutes. To be stricter because transportation subsidies are affected, they will use a 1% significance level for this check. (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value of 32 minutes, the sample size, and the observed sample result for the mean commute time to the symbols used in the solution. (C2) Using the 1% significance level, write the null and alternative hypotheses for the case where the true mean commute time is lower than 32 minutes. (C3) Classify the test and justify it in words for the case where the true mean commute time is lower than 32 minutes.

C1

- This is a population mean setting.
- Let μ be the true population mean commute time for all students in the district.
- Let μ_0 be the benchmark estimate, with $\mu_0 = 32$.

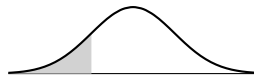
- Let n be the sample size, with $n = 40$.
- Let $\bar{x}$ be the sample mean commute time, with $\bar{x} = 34$.
- Let σ be the known population standard deviation, with $\sigma = 8$.
- Let α be the significance level, with $\alpha = 1\%$.
- The alternative claim in this problem is that the mean is lower than the benchmark.

C2

$H_0: \mu = 32$, $H_a: \mu < 32$, $\alpha = 0.01$. In words, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is lower than the benchmark.

C3

Left-tailed test because the alternative claims the parameter is lower than the benchmark.



P7 7 Cafeteria card usage

A university finance office is reviewing cafeteria payment patterns after introducing a larger card-only meal discount this term. The office is specifically concerned that card usage might now exceed the usual planning benchmark, because a higher-than-expected card share would increase transaction fees and change next semester's contract terms. The benchmark proportion of purchases paid with student cards is 70%. In a sample of 300 purchases, 198 are paid with student cards. Because the next contract decision affects a limited budget, they will use a 2% significance level for this check. (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value of 70%, the sample size, and the observed sample result for the card-payment proportion to the symbols used in the solution. (C2) Using the 2% significance level, write the null and alternative hypotheses for the case where the true card-payment proportion is higher than 70%. (C3) Classify the test and justify it in words for the case where the true card-payment proportion is higher than 70%.

C1

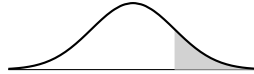
- This is a population proportion setting.
- Let p be the true population proportion of cafeteria purchases paid with student cards.
- Let p_0 be the benchmark card-payment proportion, with $p_0 = 70\%$.
- Let n be the sample size, with $n = 300$.
- Let x be the observed number of purchases paid with student cards, with $x = 198$.
- Let α be the significance level, with $\alpha = 2\%$.
- The alternative claim in this problem is that the proportion is higher than the benchmark.

C2

$H_0: p = 0.70$, $H_a: p > 0.70$, $\alpha = 0.02$. In words, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is higher than the benchmark.

C3

Right-tailed test because the alternative claims the parameter is higher than the benchmark.



P8 8 Quiz score average in a career-focused course

A department is reviewing a career-focused course after replacing several quiz items and adjusting the grading rubric midterm. Because of these changes, staff are not assuming scores would move only up or only down; they need to check whether the typical quiz score is still aligned with the long-run target. The target is an average quiz score of 68 points. A sample of 25 students has an average score of 71 points. Assume the population standard deviation is known to be 9 points. Because tutoring hours and review sessions may be expanded under stricter or more flexible policies, the department will consider three significance levels for this check: 1%, 2%, and 5%. (C1) Identify whether the claim is about a population proportion or a population mean, and match the parameter, the benchmark value of 68 points, the sample size, the observed sample result, and the three stated significance levels to the symbols used in the solution. (C2) For each stated significance level, write the null and alternative hypotheses for the case where the true mean quiz score is different from 68 points. (C3) Classify the test and justify it in words for the case where the true mean quiz score is different from 68 points, and state whether the tail type changes across the stated significance levels.

C1

- This is a population mean setting.
- Let μ be the true population mean quiz score for all students in the department.
- Let μ_0 be the benchmark target mean, with $\mu_0 = 68$.
- Let n be the sample size, with $n = 25$.
- Let $\bar{x}$ be the sample mean quiz score, with $\bar{x} = 71$.
- Let σ be the known population standard deviation, with $\sigma = 9$.
- Let α represent the significance level, and here α will be considered at 1%, 2%, and 5%.
- The alternative claim in this problem is that the mean is different from the benchmark.

C2

At $\alpha = 0.01$: $H_0: \mu = 68$, $H_a: \mu \neq 68$.

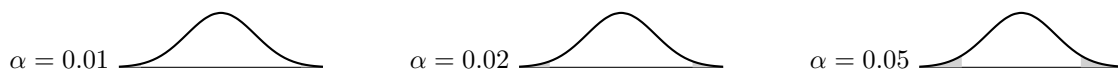
At $\alpha = 0.02$: $H_0: \mu = 68$, $H_a: \mu \neq 68$.

At $\alpha = 0.05$: $H_0: \mu = 68$, $H_a: \mu \neq 68$.

In words, for all three significance levels, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is different from the benchmark. The hypotheses stay the same; only the significance level changes.

C3

This is a two-tailed test at all three significance levels because the alternative hypothesis is $H_a: \mu \neq 68$, and tail type is determined by the alternative hypothesis. As α changes from 0.01 to 0.02 to 0.05, the test stays two-tailed, but the shaded rejection regions in both tails become wider for larger α and narrower for smaller α .



P9 9 On-time assignment submission rate after timetable changes

A college advising office adjusted weekly class timetables so that students in foundation courses have longer breaks before assignment deadlines. The office expects this change to support time management, but the concern is whether the share of on-time submissions may have dropped below the usual performance standard during the transition period. The benchmark proportion of on-time submissions is 88%. In a review sample of 250 submitted assignments, 210 are on time. Because scholarship eligibility reviews depend on this indicator, they will examine this same concern at significance levels of 1%, 5%, and 10%.

Questions/Tasks.

- (C1) Identify whether the setting is a proportion claim or a mean claim, and map the benchmark value, sample information, and the three significance levels to the correct symbols.
- (C2) Write the null and alternative hypotheses for the claim at each of the three stated significance levels.
- Use the first significance level.
 - Use the second significance level.
 - Use the third significance level.
- (C3) Classify the tail type for the claim at each of the three stated significance levels and justify it in words.

C1

- This is a population proportion setting.
- Let p be the true population proportion of on-time assignment submissions.
- Let p_0 be the benchmark on-time submission proportion, with $p_0 = 88\%$.
- Let n be the sample size, with $n = 250$.
- Let x be the observed number of on-time submissions, with $x = 210$.

- Let α represent the significance level, and here α is considered at 1%, 5%, and 10%.
- The concern in this setting is whether the true on-time submission proportion is below the 88% standard.

C2

(a) At $\alpha = 0.01$: $H_0: p = 0.88$, $H_a: p < 0.88$.

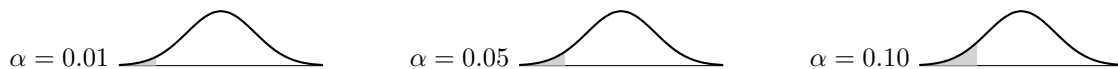
(b) At $\alpha = 0.05$: $H_0: p = 0.88$, $H_a: p < 0.88$.

(c) At $\alpha = 0.10$: $H_0: p = 0.88$, $H_a: p < 0.88$.

In words, for all three significance levels, the null hypothesis says that the population proportion equals the benchmark, while the alternative says that the population proportion is lower than the benchmark. The hypotheses stay the same; only the significance level changes.

C3

This is a left-tailed test at all three significance levels because the alternative hypothesis is $H_a: p < 0.88$, and tail type is determined by the alternative hypothesis. As α changes from 0.01 to 0.05 to 0.10, the test stays left-tailed, but the shaded rejection region in the left tail becomes wider for larger α and narrower for smaller α .



P10 10 Average customer waiting time at a campus help desk

A student services center tracks waiting times at its help desk after introducing a new digital ticketing system. Staff promised that the system would keep support lines efficient, but administrators are concerned that average waiting time may now exceed the service cap used in annual quality reports. The benchmark average waiting time is 14 minutes. A sample of 36 students shows an average waiting time of 15.8 minutes. Assume the population standard deviation is known to be 4 minutes. Because service-level targets are part of next semester's staffing plan, they will examine this same concern at significance levels of 1%, 2%, and 5%.

Questions/Tasks.

- (C1) Identify whether the setting is a proportion claim or a mean claim, and map the benchmark value, sample information, and the three significance levels to the correct symbols.
- (C2) Write the null and alternative hypotheses for the claim at each of the three stated significance levels.
- Use the first significance level.
 - Use the second significance level.
 - Use the third significance level.
- (C3) Classify the tail type for the claim at each of the three stated significance levels and justify it in words.

C1

- This is a population mean setting.
- Let μ be the true population mean waiting time for students at the help desk.
- Let μ_0 be the benchmark mean waiting time, with $\mu_0 = 14$.
- Let n be the sample size, with $n = 36$.
- Let $\bar{x}$ be the sample mean waiting time, with $\bar{x} = 15.8$.
- Let σ be the known population standard deviation, with $\sigma = 4$.
- Let α represent the significance level, and here α is considered at 1%, 2%, and 5%.
- The concern in this setting is whether the true mean waiting time is above the 14-minute service cap.

C2

(a) At $\alpha = 0.01$: $H_0: \mu = 14$, $H_a: \mu > 14$.

(b) At $\alpha = 0.02$: $H_0: \mu = 14$, $H_a: \mu > 14$.

(c) At $\alpha = 0.05$: $H_0: \mu = 14$, $H_a: \mu > 14$.

In words, for all three significance levels, the null hypothesis says that the population mean equals the benchmark, while the alternative says that the population mean is higher than the benchmark. The hypotheses stay the same; only the significance level changes.

C3

This is a right-tailed test at all three significance levels because the alternative hypothesis is $H_a: \mu > 14$, and tail type is determined by the alternative hypothesis. As α changes from 0.01 to 0.02 to 0.05, the test stays right-tailed, but the shaded rejection region in the right tail becomes wider for larger α and narrower for smaller α .

